**STOCHASTIC METHODS IN FINANCE**
**STAT0013**
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# Exercises 5 - Stochastic Calculus

1. Find the stochastic differential equation (SDE) satisfied by the square of a stock price that follows geometric Brownian motion. What is this process?

2. (a) Suppose that $g(t)$ is a *deterministic* differentiable function for $t > 0$, with $g(0) = 0$. Show that a solution to the ordinary differential equation

$$dx(t) = ax(t)dt + bx(t)dg(t)$$

with boundary condition $x(0) = x_0 \neq 0$ is

$$x(t) = x_0 e^{at+bg(t)}.$$

*Hint:* Write $dg(t)$ as $g'(t)dt$ and then use the variable separation technique and take integrals on both sides of the equation.

(b) Show that the process $X_t = x\, e^{at+bW_t}$, where $W_t$ is standard Brownian motion, satisfies the SDE

$$dX_t = \big(a + \tfrac{b^2}{2}\big) X_t\, dt + bX_t\, dW_t$$

with initial condition $X_0 = x$.

3. Show that the Itô process $X_t = e^{W_t} e^{-t/2}$ (with $W_t$ a standard Brownian motion) satisfies the stochastic differential equation

$$dX_t = X_t\, dW_t.$$

4. A zero-coupon government bond pays £100 at time $T$, and has price denoted by $B_t$. In the course so far we have assumed that the risk-free rate is constant and deterministic. In more advanced models, the risk-free rate can be modelled itself as a stochastic process. It has been

suggested that the short-term interest rate, $r_t$, will not be constant over time but will in fact follow the stochastic process

$$dr_t = a(b - r_t)dt + c\, r_t\, dz_t$$

where $a$, $b$, $c$ are positive constants and $z_t$ is a standard Brownian motion. Under this assumption, derive the SDE for the government bond price $B_t$ for $t < T$.